BTC Risk Assessment: Analyzing Trends in Prices and Fraudulent Activity

Will Park - wtpark@syr.edu

CIS 400 - Data Science in Practice

May 2, 2025

Abstract:

This project explores the interplay between public interest in Bitcoin, its market price trends, and the prevalence of cryptocurrency-related fraud. Using data from Google Trends, Kaggle, and various fraud datasets, this project blended exploratory and descriptive research methods to identify correlations and forecast outcomes. Key outputs include a multivariate regression used to predict Bitcoin prices with general risk boundaries. Exploratory analysis of fraud patterns were used comprehensively to create a forecasting model with interpretable results. However, fraud prediction was ultimately inconclusive, suggesting the need for a different, more advanced modeling method.

<u>Data collection</u>: public interest related data was sourced from Google Trends while Bitcoin price, fraud, and fraud frequency data were sourced from Kaggle.

<u>Libraries utilized</u>:

- readr: used for importing CSV datasets.
- dplyr: used for data wrangling (filtering, grouping, summarizing, etc).
- lubridate: used to extract and manipulate specific date/time information from large datasets spanning back over a decade.
- ggplot2: used to create visualizations like line charts, scatter plots, and bar charts.
- scales: enhanced plot formatting on axes and enhanced readability.
- tidyverse: used to load several packages simultaneously.

Technique Classification:

1. Exploratory Data Analysis:
    - Time-series plots of Bitcoin price and Google Trends were used to explore patterns and trends. Fraud cases were also summarized by year and type in order to further break them down into a multi-dimensional analysis.
2. Descriptive Data Analysis:
    - Yearly averages were aggregated to show the price changes in Bitcoin since its inception. Fraud trends and frequencies were also explored.
3. Predictive modeling:
    - A linear regression was created to forecast Bitcoin price behavior using public interest. The year was added as a covariate for a better forecasting model.

Introduction:

Bitcoin, the most prominent cryptocurrency, has experienced explosive growth in both market capitalization and public awareness over the past decade. As digital assets have become more integrated into mainstream finance, their volatility and association with cybercrime have raised significant concerns for consumers, regulators, and investors alike. The decentralized and pseudonymous nature of cryptocurrencies introduces both opportunity and risk—offering financial freedom while also enabling fraudulent activities with minimal traceability.

This project seeks to examine the relationship between three core elements of the Bitcoin ecosystem: public interest (as measured by Google search trends), historical pricing behavior, and the frequency and types of fraud within the cryptocurrency space. By combining data visualization with basic predictive modeling, the project aims to identify patterns and assess whether consumer sentiment and historical data can help forecast price fluctuations or signal periods of heightened fraud risk.

Background:

Public sentiment has long been a driver of speculative asset prices. In the case of Bitcoin, search interest frequently spikes alongside major price swings, suggesting a strong behavioral finance component to market movement. Google Trends data provides a unique lens into retail investor attention, which may correlate with market sentiment and ultimately influence price behavior.

Simultaneously, the anonymity and decentralized structure of blockchain systems have made Bitcoin a prime target for illicit activities, including ransomware, phishing, and theft. The BitcoinHeist dataset—compiled from tagged fraudulent wallet addresses—provides a rich foundation for studying how fraud has evolved over time and which transaction features may serve as warning signs.

While prior research has often explored Bitcoin pricing or fraud in isolation, few projects examine how these components interact. This project aims to fill that gap by integrating trend data, market data, and fraud metadata to build a comprehensive view of Bitcoin-related risk.

Problem Description:

Despite a growing body of research into Bitcoin pricing and blockchain fraud, most analyses examine these elements in isolation. A gap exists in unified models that jointly analyze consumer sentiment, price trends, and fraudulent activity. Understanding these dynamics in combination could offer useful insights for risk mitigation, fraud prevention, and market behavior forecasting.

Problem Statement:

Can historical pricing data and public interest trends help forecast future Bitcoin prices? Additionally, can network-based features from fraudulent address data predict the likelihood of fraud? This project investigates both questions using exploratory analysis and statistical modeling.

Aim & Objectives:

To analyze the relationships between Bitcoin market behavior, public interest trends, and cryptocurrency fraud using exploratory and predictive methods.

1. Visualize historical Bitcoin prices and Google Trends search interest from 2015–2025.

2. Develop regression models to predict Bitcoin price based on public interest and time.

3. Analyze fraud trends over time and identify the most common fraud types in Bitcoin transactions.

4. Construct a binary classification model using network transaction features to predict fraudulent activity.

5. Evaluate the effectiveness and limitations of the models and identify areas for improvement.

Main section:

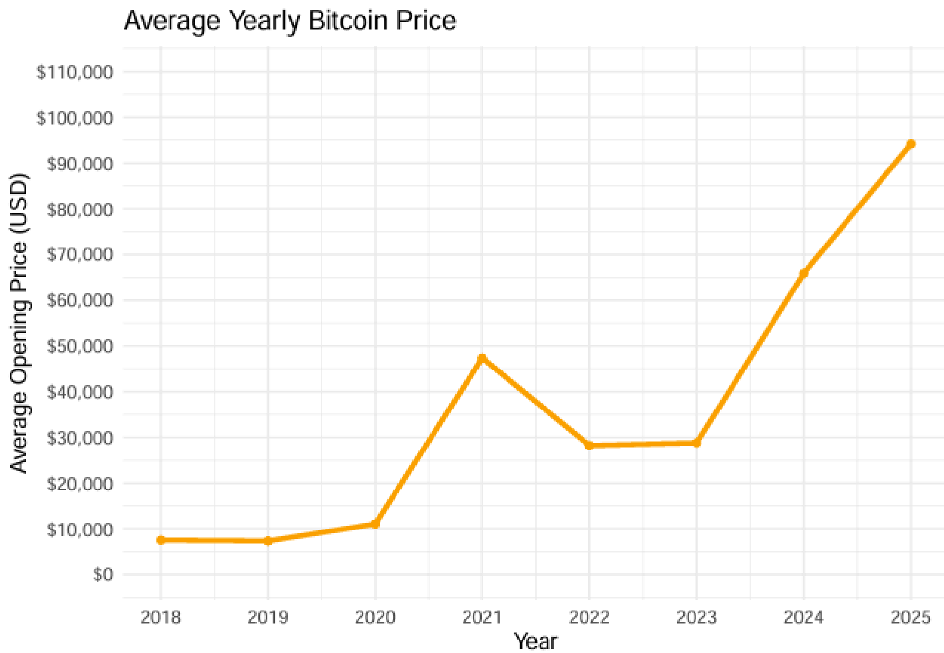

Figure 1

To fully understand the scope of my project, I aggregated daily opening price data over the past 7 years into yearly averages using a historical price dataset from Kaggle. Timestamp fields were converted into date objects before the year was extracted, the data was then filtered to the target range and summarized toe compute the mean opening price each year. Figure 1 reveals several distinct phases in Bitcoin's price evolution. The cryptocurrency maintained a relatively stable range from 2018-2020 before seeing a dramatic surge in 2021 as pandemic-era speculation. A corrective decline ensued during 2022-2023 with a rebound beginning in later 2024 in conjunction with the most recent election cycle.

By converting data into yearly trends this visualization acts as an integral exploratory tool. It highlights Bitcoin's volatile nature and enables temporal comparisons that inform subsequent regression modeling. It provides a contextual infrastructure for forecasting methods explored in later sections while simultaneously reinforcing the importance of long0term price behavior in shaping predictive models.

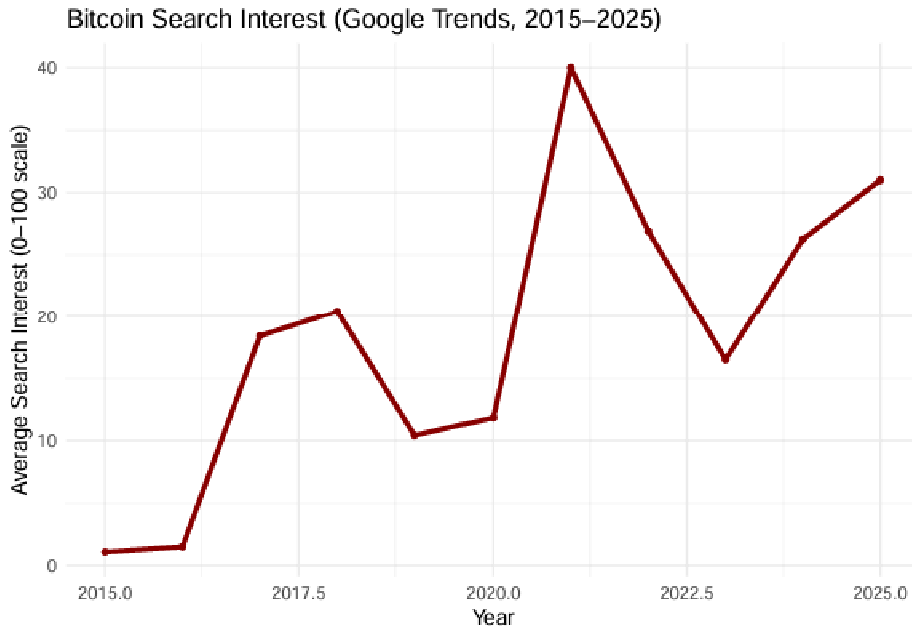

Figure 2.

Figure 2 is a visualization that captures trends in public sentiment towards Bitcoin. Annual search interest was gathered by filtering terms related to Bitcoin - “Bitcoin price”, “Bitcoin trends”, “buy or sell Bitcoin”, etc - before being exported as a CSV file. This included a normalized index of search frequency from 0-100 and was structured with year along with queries. The line plot highlights several spikes in public interest, most notably around 2017-2018, 2021, and 2025, aligning with major price surges, speculative media coverage, and politics. There is a drop following these peaks which is consistent with post-boom market corrections. Public attention gradually rises after 2023, potentially reflecting renewed adoption and some market recovery as the price changed. The visualization is an effective showing of cyclical interest patterns which generally correlate with price volatility and form the basis for predictive modeling in subsequent sections.

By quantifying sentiment over time and establishing a link to known market events, the visualization supports the objective of tracking and interpreting long-term trends in public interests. This also effectively established a data-driven rationale for including search behavior in the multivariate regression used later on.

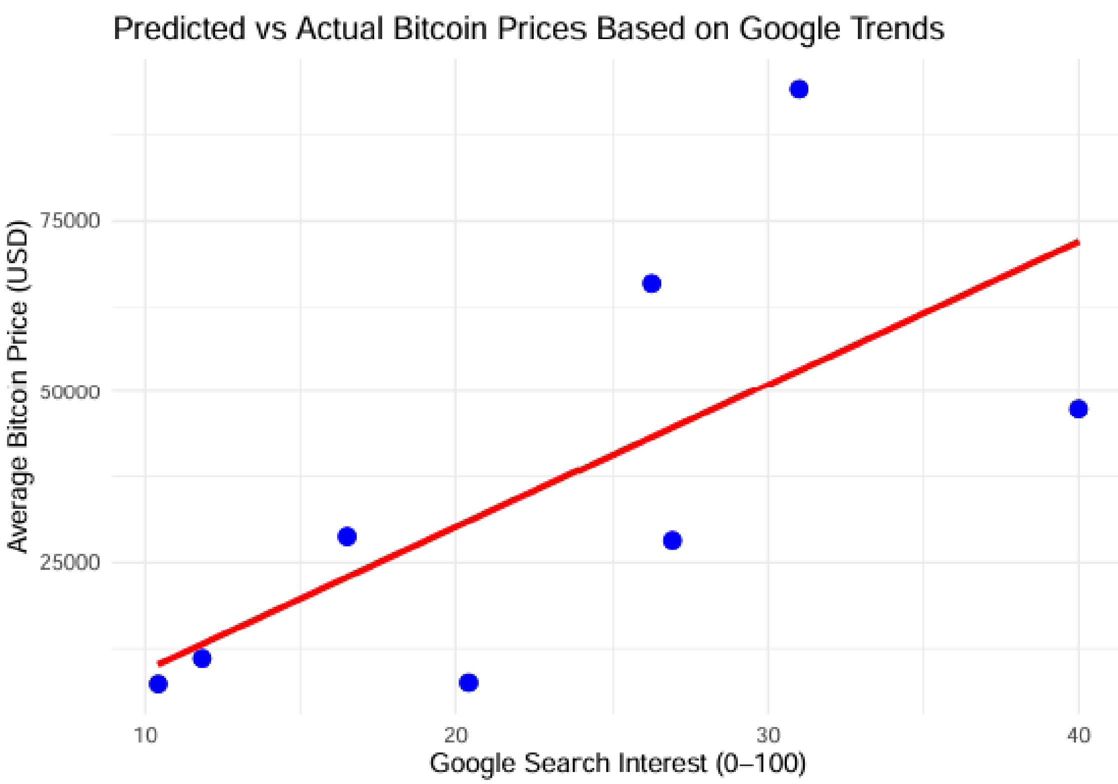

Figure 3.

To assess whether public interest can predict Bitcoin's price, a linear regression model was developed using yearly average search interest from Google Trends and corresponding Bitcoin prices. The datasets trend_data and btc_yearly_avg were merged on the Year column, and the model was fit using the formula Avg_Open ~ bitcoin.price.searches. Figure 3 compares actual prices (blue points) to the predicted regression line (red). While the model shows a generally upward trend—indicating a positive relationship between search volume and price—the scatter of residuals reveals considerable variance, particularly at high-interest levels.

The model produced an R^2 of 0.458 with a p-value near 0.065, suggesting moderate predictive strength but not statistical significance at the 5% threshold. These results indicate that search interest is somewhat correlated with price behavior, though not strongly enough on its own to produce reliable forecasts without additional variables.

This task models and evaluates the relationship between public sentiment and Bitcoin pricing. It directly supports the objective of developing a regression model to forecast Bitcoin prices using sentiment data. Although the linear model suggests a moderate positive correlation, its limited statistical significance indicates that more complex/multivariate approaches are needed to refine predictive reliability.

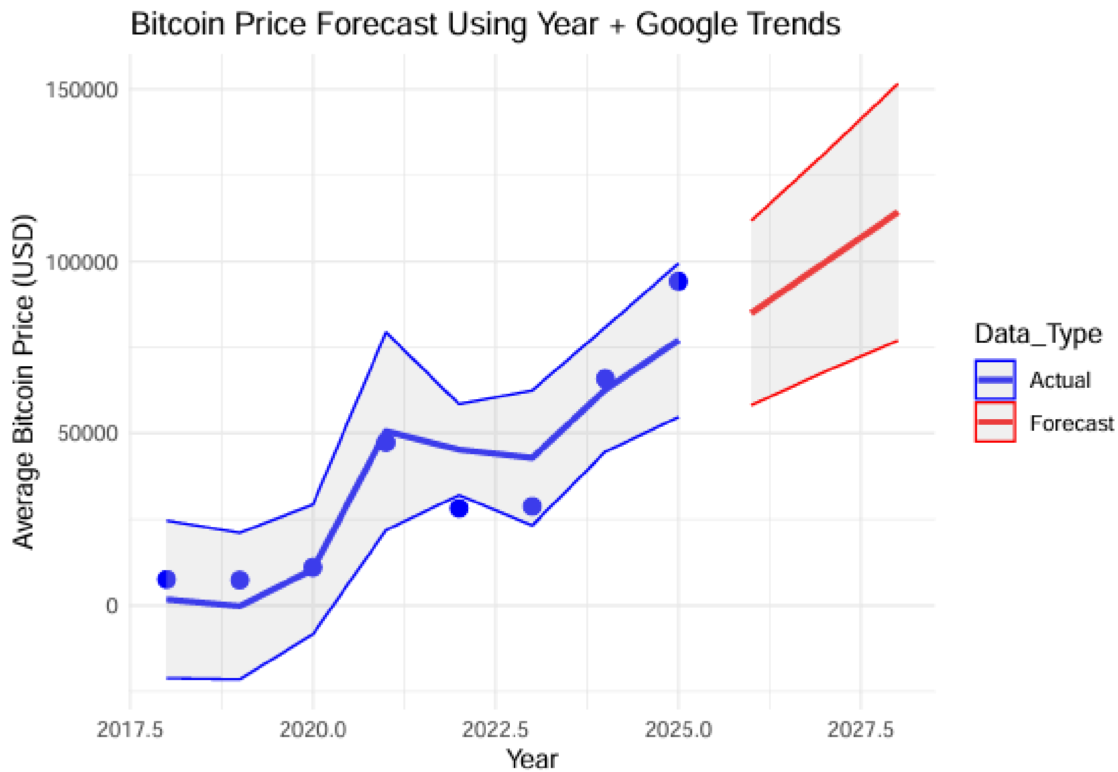

Figure 4.

To improve upon the limitations of the simple linear model, a multivariate linear regression was implemented using both Google search interest and year as predictors of Bitcoin's average price. The training dataset included observed values from 2018 to 2025, and future projections were generated manually for 2026–2028 using hypothetical interest values. As shown in Figure 4, the model was able to better capture the upward market trajectory while producing forecast intervals that reflect uncertainty. Actual prices are represented in blue, with fitted trend lines and confidence bands, while forecasted values are shown in red with extended intervals. By incorporating the time variable, the model captures the compounding effect of both market momentum and sentiment over time. Intervals between the forecast and actual price line depict a reasonable area of BTC price projection.

While the projection suggests continued growth, the widening bands beyond 2025 highlight the model's growing uncertainty in long-term predictions. Overall, the multivariate approach offers a more nuanced and interpretable forecasting method than relying on sentiment data alone.Given the time of this data being collected (April 2025), this model is unfortunately not a reliable forecast. Following the new United States presidential administration taking office, the cryptocurrency economy has become volatile, inhibiting the accuracy of this forecast.

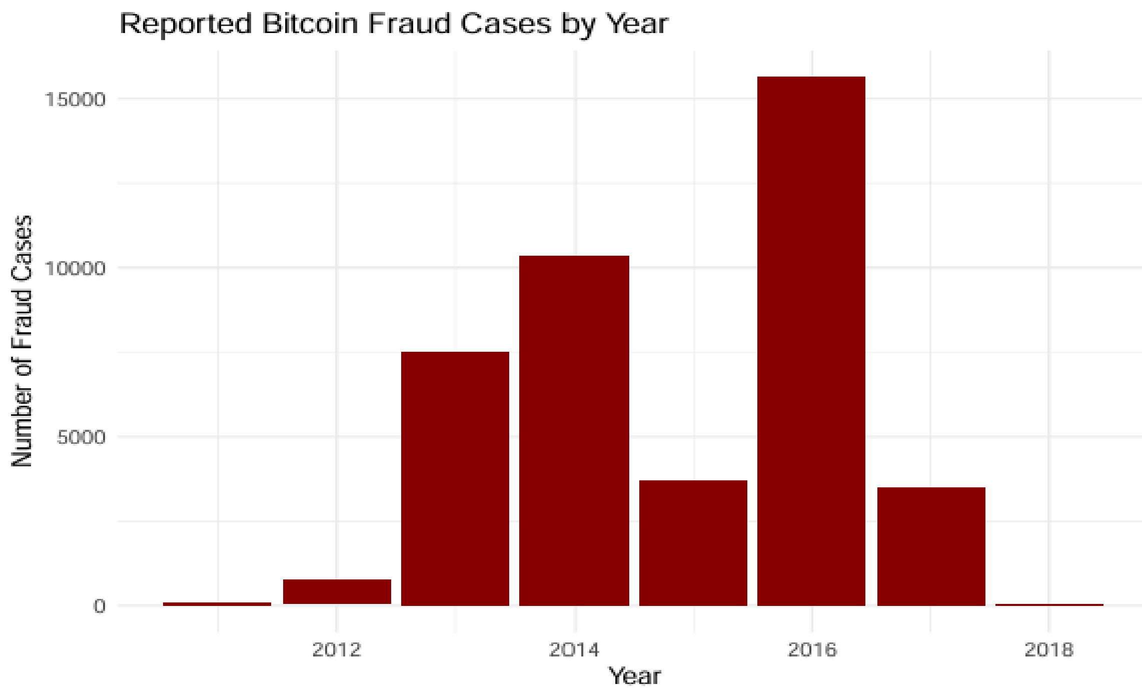

Figure 5.

This bar chart was created using the BitcoinHeist dataset as a fallback due to a failed API call to CryptoScamDB. While attempting to locate a more current and complete dataset, I found that many public repositories on recent crypto fraud activity—such as Chainalysis and DFPI scam trackers—either restrict access behind paywalls, require organizational affiliation, or provide limited structured data for academic use. Fortunately, the BitcoinHeist dataset provided ample historical information to analyze fraudulent activity in the Bitcoin ecosystem. It includes labeled transaction data from 2011 to 2018, allowing for a breakdown of fraud incidents by year. During preprocessing, legitimate addresses (label == "white") were filtered out, and remaining cases were grouped by year and counted. The resulting bar chart reveals a significant spike in activity between 2013 and 2016, with a peak of over 15,000 reported cases in 2016. This period aligns with the broader adoption of ransomware and phishing tactics that exploited early gaps in blockchain awareness and security. However, it is important to note that the data ends in 2018 and may not reflect recent developments in fraud patterns, scam typologies, or DeFi-related exploits that have since emerged. It is plausible to assume that recent advancements in mining and security have reduced fraud cases in the past 7 years.

By investigating and interpreting historical fraud activity across time, this visualization supports the objective of analyzing trends in fraud to identify periods of heightened risk within the Bitcoin ecosystem. Unfortunately, the limited scope highlights the need for more recent and comprehensive data to create a more accurate fraud pattern assessment.

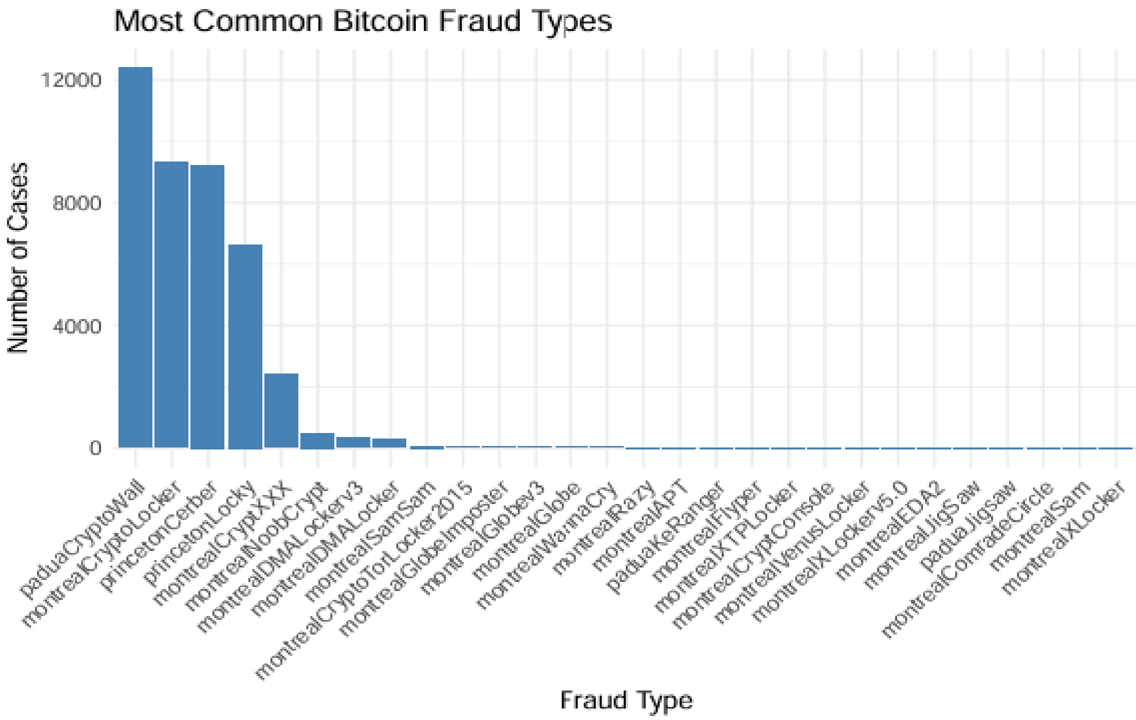

Figure 6.

To better understand the nature of fraudulent activity within the Bitcoin ecosystem, I analyzed the frequency distribution of different fraud types using the BitcoinHeist dataset. After filtering out legitimate transactions (label == "white"), the remaining entries were grouped by their fraud label and counted. The resulting bar chart (Figure 6) reveals a highly skewed distribution, where just a few fraud types account for the vast majority of reported cases. Notably, PaduaCryptoWall, MontrealCryptoLocker, and PrincetonCerber dominate the dataset—each representing well-known ransomware strains linked to phishing-based attacks.

What makes this particularly interesting is the disproportionate concentration of fraud activity in only a handful of tactics, despite the technical diversity of the broader crypto landscape. While dozens of other scam types exist, most appear only a few times, forming a long tail of low-frequency attacks. This concentration of risk reinforces the need for targeted prevention strategies that focus on the most disruptive and persistent fraud mechanisms. It also suggests that fraud in crypto markets may evolve more through repetition of proven exploits than through diversification, making pattern recognition and early detection both possible and essential.

This analysis of fraud composition supports the same objective as Figure 5 and identifies dominant fraud types and trends. This reveals patterns that can inform risk modeling and prioritization of defense strategies.

Conclusion & Further Work:

This project examined the relationship between public interest in Bitcoin, its historical pricing behavior, and the prevalence of cryptocurrency-related fraud. Using datasets from Google Trends, Kaggle, and the BitcoinHeist dataset, a combination of exploratory data visualization and predictive modeling techniques was applied to uncover patterns across market sentiment, price trends, and fraud dynamics. Linear and multivariate regression models revealed a moderate correlation between Google search interest and Bitcoin price, particularly when time was incorporated as a predictor. Visualizations captured key periods of speculative growth, price corrections, and recoveries between 2018 and 2025, helping to contextualize market cycles with sentiment fluctuations.

However, attempts to predict fraud using network-level wallet features from the BitcoinHeist dataset were largely unsuccessful. The logistic regression model failed to identify statistically significant predictors of fraud activity, likely due to limitations in the dataset—such as its relatively small feature set, imbalance between fraudulent and non-fraudulent cases, and cutoff year of 2018. These constraints hindered the model's generalizability and reduced its relevance to the rapidly evolving landscape of crypto crime. Furthermore, the binary labeling system ("fraud" vs. "white") oversimplified the nuanced nature of scam behavior, preventing deeper classification of fraud type or severity.

To improve future research, several enhancements should be pursued. First, expanding the dataset to include more recent fraud activity—possibly through partnerships with platforms like Chainalysis, the DFPI Scam Tracker, or public crypto crime aggregators—would significantly improve the model's temporal relevance. Second, alternative machine learning techniques such as decision trees, random forests, or neural networks should be explored to handle class imbalance and non-linear relationships. These models could better capture subtle feature interactions that traditional regression overlooks. Additionally, incorporating alternative data sources—such as blockchain network structures, transaction patterns, wallet metadata, or social media sentiment—may yield stronger predictors for both fraud risk and price volatility.

Ultimately, these improvements could enable the development of more robust early-warning systems and risk monitoring frameworks, equipping exchanges, regulators, and investors with timely, data-driven insights into both market dynamics and security threats in the cryptocurrency sector.

Revised Proposal

Introduction:

Over the past decade, Bitcoin has emerged as the dominant cryptocurrency in both market capitalization and public discourse. Its rise has been accompanied by intense market speculation, public attention, and a growing ecosystem of fraud and cybercrime. This project investigates how Bitcoin's price and associated fraud patterns have evolved in relation to public sentiment—measured using Google search interest—and whether these elements can be used to build predictive models. Rather than analyzing the broader crypto ecosystem, this study focuses exclusively on Bitcoin to maintain analytical clarity and align with available datasets.

Subject Area:

This project centers on three primary elements: Bitcoin market price trends, Google search interest, and cryptocurrency-related fraud incidents. By examining these factors in combination, the project explores both market behavior and systemic vulnerabilities within the Bitcoin ecosystem.

Problem:

While many studies have explored Bitcoin prices or cryptocurrency fraud independently, fewer have considered the interaction between public sentiment, market volatility, and fraud risk. There remains a gap in understanding how these dimensions overlap and whether they can be used together for forecasting price behavior or detecting early signs of fraud.

Approach:

Data was collected from public sources and pre-cleaned CSV files:

Market Data: Bitcoin historical pricing (2018–2025) via Kaggle (btc_historical.csv)

Public Sentiment: Google Trends data (2015–2025) on the search term "Bitcoin"

Fraud Data: BitcoinHeist dataset containing address-level fraud labels (2011–2018)

The analysis proceeded in three phases:

- **Exploratory Visualizations**: Time-series plots of Bitcoin price and Google search interest; bar charts to show frequency of fraud by year and by fraud type.

- **Descriptive Analysis**: Aggregated yearly averages to observe long-term patterns and identify major shifts.

- **Predictive Modeling**:

- Linear regression to model Bitcoin price based on search interest
- Multivariate regression including year as an additional predictor
- Logistic regression to model fraud probability using address-level features (length, weight, and growth rate)

Objectives:

1. Visualize historical Bitcoin price trends and public interest from 2015–2025
2. Develop regression models to predict Bitcoin price using sentiment and time
3. Analyze fraud activity over time and identify dominant fraud types
4. Attempt fraud prediction using network features and logistic regression
5. Evaluate model performance and identify limitations or areas for improvement

Data Sources Used:

- Kaggle: btc_historical.csv and BitcoinHeistData.csv
- Google Trends: Yearly normalized interest in the search term "Bitcoin"

Note: Originally proposed APIs such as CoinGecko, Chainalysis, and IC3 were not used due to access limitations. The scope was narrowed to Bitcoin-specific, pre-existing datasets for consistency and feasibility.

Deliverables:

- Data visualizations capturing long-term trends in price, sentiment, and fraud
- Regression models and their evaluation plots
- Fraud breakdowns by type and year
- Final report outlining methods, findings, limitations, and recommendations

Aim:

To quantitatively analyze Bitcoin's market behavior, public sentiment, and fraud patterns to identify correlations, develop basic forecasting models, and evaluate risk signals within the cryptocurrency ecosystem.

References:

File used: btc_historical.csv
Kaggle. (n.d.). *Bitcoin Historical Data*. Retrieved from https://www.kaggle.com/datasets/mczielinski/bitcoin-historical-data

File used: BTCsent.csv
Google Trends. (2025). *Search interest for "Bitcoin" (2015–2025)*. Retrieved from https://trends.google.com/

File used: BitcoinHeistData.csv
Janos, T., & Tapamo, J. R. (2019). *BitcoinHeist ransomware address dataset*. UCI Machine Learning Repository. Retrieved from https://www.kaggle.com/datasets/datasnaek/bitcoin-heist-ransomware-address-dataset